

The Representer Theorem for Kernel Based Support Vector Machines

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Introduction

I will outline a proof of the representer theorem for reproducing kernel Hilbert spaces over $\mathbb{R}$, and then will use it in conjunction with the Support Vector Machine methods I have developed thus far to create a kernel based support vector machine.

Representer Theorem

Lemma 1: A Hilbert space $\mathcal{H}$ of functions from a set X to $\mathbb{R}$ (or $\mathbb{C}$) is a reproducing kernel Hilbert space if and only if $\exists K_x \in \mathcal{H}$ such that $\forall f \in \mathcal{H} \langle f, K_x \rangle = f(x)$.

Proof. $\implies$ By definition a Hilbert space H is a reproducing kernel Hilbert space if $\forall x \in X$ the linear functional $L_x : \mathcal{H} \rightarrow \mathbb{R}$ given by $L_x(f) = f(x)$ is continuous. Thus L_x is bounded on $\mathcal{H}$ as continuity implies boundedness, so we can apply Riesz's representation theorem to find a unique $g_x \in \mathcal{H}$ such that $L_x(f) = \langle f, g_x \rangle$. Setting $K_x = g_x$, it is clear that K_x is well defined as g_x is unique by Riesz's and $\langle f, K_x \rangle = \langle f, g_x \rangle = f(x)$. Thus, K_x is a reproducing kernel of $\mathcal{H}$.

$\Leftarrow$ Suppose there exists a $K_x \in \mathcal{H}$ such that $\langle f, K_x \rangle = f(x)$. Then we can define a linear functional $L_x : X \rightarrow \mathbb{R}$ $L_x(f) = \langle f, K_x \rangle$. By Cauchy-Schwarz, $|\langle f, K_x \rangle| \leq \|f\|_{\mathcal{H}}^2 \|K_x\|_{\mathcal{H}}^2$. Thus, $L_x(f)$ is bounded. Now we must show that L_x is continuous. Given $\epsilon > 0$ pick $\delta = \frac{\epsilon}{\|K_x\|_{\mathcal{H}}^2}$. Then for $g \in \mathcal{H}$ such that $\|f - g\|_{\mathcal{H}}^2 < \delta$ we have:

$$\begin{aligned} \|L_x(f) - L_x(g)\| &= |L_x(f - g)| \text{ as } L_x \text{ is a linear functional} \\ &= |\langle f - g, K_x \rangle| \\ \text{By Cauchy Scharwz we have:} \\ &\leq \|f - g\|_{\mathcal{H}}^2 \|K_x\|_{\mathcal{H}}^2 \\ &< \frac{\epsilon}{\|K_x\|_{\mathcal{H}}^2} \|K_x\|_{\mathcal{H}}^2 \\ &= \epsilon \end{aligned}$$

Thus, L_x is continuous and H is a reproducing kernel Hilbert space as required. $\square$

Lemma 2: We can characterize K in the above lemma as a function $K : X \times X \rightarrow \mathbb{R}$ by $K(x, y) = \langle K_x, K_y \rangle$.

Proof. By Lemma 1, there exists $K_y \in \mathcal{H}$ such that $\forall f \in \mathcal{H} \langle f, K_y \rangle = f(y)$. If we consider $K_x \in \mathcal{H}$, we can apply this characterization to it. We find that $K_x(y) = \langle K_x, K_y \rangle$. Thus we can define $K(x, y) = \langle K_x, K_y \rangle$ as a function from $X \times X$ to $\mathbb{R}$. In particular, this implies that K is symmetric. $\square$

Theorem: Let $\mathcal{H}$ be a reproducing kernel Hilbert space of functions over $\mathbb{R}$ with domain V , a finite dimensional vector space over $\mathbb{R}$ and a set of inputs $X = \{x^{(1)}, \dots, x^{(m)}\} \subset V$ and outputs $Y = \{y^{(1)}, \dots, y^{(m)}\} \subset \mathbb{R}$. Now define $J : \mathcal{H} \rightarrow \mathbb{R}$ by $J(f) = \sum_{i=1}^m L(f(x^{(i)}), y^{(i)}) + \lambda \|f\|_{\mathcal{H}}^2$ be a regularized risk function with respect to f . Then any function f^* that minimizes J has the form $f^* = \sum_{i=1}^m \alpha_i K(x, x^{(i)})$.

Proof. Given the H a reproducing kernel Hilbert space with kernel K , it suffices to show that $\forall f \in \mathcal{H}$ we can find $f^* = \sum_{i=1}^m \alpha_i K(x, x^{(i)})$ such that $J(f) \geq J(f^*)$. Consider the subspace $W = \text{Span}\{K(\cdot, x^{(i)})\}_{i=1}^m$. W is finite dimensional by construction, so we may write $\mathcal{H} = W \oplus W^\perp$. Thus given $f \in \mathcal{H}$, we can find $f_1 \in W$ and $f_2 \in W^\perp$ such that $f = f_1 + f_2$. Applying the reproducing property of K to f_2 on $x^{(i)}$ we see that:

$$f_2(x^{(i)}) = \langle f_2, K_{x^{(i)}} \rangle$$

By Lemma 2:

$$= \langle f_2, K(x^{(i)}, \cdot) \rangle$$

$$= \langle f_2, K(\cdot, x^{(i)}) \rangle$$

Since $f_2 \in W^\perp$:

$$= 0$$

Thus, $f(x_i) = f_1(x_i)$. Additionally by the pythagoras theorem: $\|f\|_{\mathcal{H}}^2 = \|f_1\|_{\mathcal{H}}^2 + \|f_2\|_{\mathcal{H}}^2 \geq \|f_1\|_{\mathcal{H}}^2$. Now setting $f^* = f_1$. It is clear that:

$$\begin{aligned} J(f) &= \sum_{i=1}^m L(f(x^{(i)}), y^{(i)}) + \lambda \|f\|_{\mathcal{H}}^2 \\ &\geq \sum_{i=1}^m L(f^*(x^{(i)}), y^{(i)}) + \lambda \|f^*\|_{\mathcal{H}}^2 \\ &= J(f^*) \end{aligned}$$

$\square$

Note: We assume L is convex in its first argument and lower semi-continuous to ensure it attains a minimizer in $\mathcal{H}$